ZHOU Junbo
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Education Background

Southwest Jiaotong University (Project 211, Double First-Class Initiative) **Sep. 2023 – Today**
Doctor of Electrical Engineering Chengdu, China

- Supervisor: Prof. Dr. Guoqiang Gao
- Research Direction: Hyperspectral image processing in electrical engineering by using deep learning methods

Wuhan Polytechnic University **Sep. 2020 – Jun. 2023**
Master of Electronic and Information Engineering Wuhan, China

- GPA: 87.06/100
- Supervisor: Prof. Dr. Shan Zeng
- Research Direction: Hyperspectral image processing and deep learning methods

Tiangong University (Double First-Class Initiative) **Sep. 2016 – Jun. 2020**
Bachelor of Information and Computational Science Tianjin, China

- GPA: 85.68/100

Publications

Journal Publications

- **J. Zhou**, S. Zeng, G. Gao, et al. "A Novel Spatial–Spectral Pyramid Network for Hyperspectral Image Classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1-14, 2023, Art no. 5519314, DOI: 10.1109/TGRS.2023.3303338. **(SCI Q1, IF = 9.4)**
- **J. Zhou**, G. Gao, Y. Guo, et al, "Fast contamination recognition network: a knowledge distillation based hyperspectral outdoor insulator contamination level recognition network," *Expert Systems with Applications*, vol. 300, p. 130401, 2025. DOI: 10.1016/j.eswa.2025.130401. **(SCI Q1, IF = 9.4)**
- **J. Zhou**, S. Zeng, Z. Xiao, et al., "An enhanced spectral fusion 3D CNN model for hyperspectral image classification," *Remote Sensing*, vol. 14, no. 21, p. 5334, 2022. DOI: 10.3390/rs14215334. **(SCI Q1, IF = 4.3)**
- Y. Fan, Y. Guo, Y. Liu, S. Xiao, **J. Zhou**, et al., "A pixel-level assessment method of the aging status of silicone rubber insulators based on hyperspectral imaging technology and IPCA-SVM model," *Expert Systems with Applications*, vol. 263, Art. no. 125788, 2024. DOI: 10.1016/j.eswa.2024.125788. **(SCI Q1, IF = 9.4)**
- Y. Liu, Y. Guo, Y. Fan, **J. Zhou**, et al., "Optical imaging technology application in transmission line insulator monitoring: A review," *IEEE Transactions on Dielectrics and Electrical Insulation*, vol. 31, no. 6, pp. 3120–3132, 2024. DOI: 10.1109/TDEI.2024.3351093. **(SCI Q1, IF = 3.2)**
- H. Li, Y. Ou, Y. Guo, Y. Fan, **J. Zhou**, et al., "A Hyperspectral Method for Joint Assessment of Aging and Contamination States in Composite Insulators," *IEEE Transactions on Dielectrics and Electrical Insulation*, 2026. DOI: 10.1109/TDEI.2026.3712704. **(SCI Q1, IF = 3.2)**

Conference Paper

- **J. Zhou**, G. Gao, C. Zhang, et al., "HSRNet: Hyperspectral Retentive Network for Hyperspectral Image Classification," in *IEEE International Joint Conference on Neural Networks (IJCNN)*, 2026. **(Accepted)**

Research Experience

Doctoral Research Project **Sep. 2023 – Today**

Research Topic: *Hyperspectral Imaging-Based Insulator Contamination Characterization and Recognition under Natural Illumination and Variable Observation Distances*

Responsibilities:

- Designed outdoor hyperspectral experiments and datasets for insulator contamination under varying illumination, weather, and observation-distance conditions.
- Developed deep learning methods for spectral-spatial representation, contamination-level recognition, hyperspectral unmixing, and lightweight inference; conducted model implementation and evaluation.

Master Research Project **Sep. 2020 – Jun. 2023**

Research Topic: *Hyperspectral Image Classification Using 3D Convolutional Neural Networks*
Responsibilities:

- Designed 3D CNN-based classification models integrating spectral attention, multi-scale convolutions, and spatial-spectral feature fusion.
- Implemented the models and conducted ablation, parameter, and comparative experiments on multiple benchmark datasets; prepared the resulting journal publications.

National Key Research and Development Program of China Project **Jun. 2025 – Today**

Grant No.: 2024YFB2606300 | Project Fund: RMB 90.5 million | Project Leader: Prof. Guangning Wu | Position: Project Core Member

Research Topic: *Interaction Characteristics and Analysis Methods for Electricity, Hydrogen, and Thermal Multi-Energy Flows*
Responsibilities:

- Contributed to proposal preparation and technical roadmap design for multi-energy-flow modeling, simulation, and interaction analysis in transportation hubs.
- Developing coupled models of electricity, hydrogen, and thermal energy flows and data-driven methods to characterize energy transmission, conversion, and multi-timescale interactions.

Academic Service

- Peer Reviewer: *IEEE Transactions on Geoscience and Remote Sensing*, *Engineering Applications of Artificial Intelligence*, *Pattern Recognition*, *Scientific Data*, and other international journals.

Certificates & Awards

- Yanghua Emerging Talent Scholarship, Southwest Jiaotong University, 2023
- Outstanding Master's Thesis Award, Wuhan Polytechnic University, 2023
- Outstanding Graduate Student, Wuhan Polytechnic University, 2023
- Third Prize, 19th China Post-Graduate Mathematical Contest in Modeling ("Huawei Cup"), 2022
- Outstanding Graduate Student Award, Wuhan Association for Computer Software Engineering, 2022
- Yinianyuan Scholarship and Outstanding Graduate Student Role Model, Wuhan Polytechnic University, 2021-2022
- Third Prize, 18th China Post-Graduate Mathematical Contest in Modeling ("Huawei Cup"), 2021
- Second Prize, 15th Graduate Zhixing Forum, Wuhan Textile University, 2021

Skills

- Language: Mandarin (Native), English (CET-6)
- Substantial skills of programming languages and statistics software including Python, Linux, MS Office, Adobe Illustrator, Photoshop